Paytm Saathi

A voice-first, vernacular support teammate for small merchants

Product Requirements Document | Paytm Build for India AI Hackathon | Track 3: Autonomous AI Teammates (customer service)

Status: working draft for Stage 1 (pitch deck) and Round 2 planning. **Working name:** "Saathi" (rename freely). **Important honesty note:** Paytm's internal APIs and data are not public, so the prototype uses **mocked services and synthetic data**. All metrics below are **proposed targets to validate**, not measured results. Add sourced market figures yourself before presenting.

1. Summary

Small merchants (kirana shops, street vendors, tea stalls) depend on Paytm QR codes and Soundbox devices to get paid. When a device goes silent or a settlement does not arrive, the merchant loses sales and trust, and today's fix is a call centre, a queue, and repeated questions. **Saathi** is an AI teammate that takes a merchant's **voice note in their own language**, diagnoses the issue from device and settlement data, **acts** (remote fix, retry, technician booking), and **replies by voice**. Risky or unclear cases go to a human agent with a ready summary.

2. Problem statement

A merchant whose Soundbox stops announcing payments, or whose settlement is delayed, cannot easily explain or resolve it: they may not type comfortably, may not navigate app menus, and cannot afford to wait on hold during business hours. Support agents then repeat the same diagnostic steps (check device status, check settlement, check bank-side delay) for every ticket.

- **Who suffers:** small merchants (lost sales, uncertainty about whether money arrived) and support teams (repetitive, rule-based work).
- **Why now:** Indic speech models and tool-calling LLMs are mature enough to run this loop end to end.
- **Evidence to add:** one or two sourced figures (merchant count, Soundbox base, support ticket share for settlement/device issues). Do not invent numbers.

3. Users and personas

Persona	Needs	Pain today
Merchant (primary): small shop owner, regional-language speaker, voice-comfortable, limited typing	Know quickly if money arrived; get the device working; speak, not type	Hold times, menu navigation, language barrier, lost sales during downtime
Support agent (secondary)	Receive escalations with full context; approve medium-risk actions in one click	Re-reading history, repeating diagnostic steps
Field technician (tertiary)	Visit only when needed, with the diagnosis already attached	Wasted visits for issues fixable remotely

4. Goals, non-goals and success metrics

Goals

- Resolve common merchant device and settlement issues without a human, within clear policy limits.
- Make support accessible by voice in the merchant's language (Gujarati and Hindi first).

- Hand off to humans with context, never leaving the merchant stuck.

Non-goals (for the hackathon)

- Real-time phone-call voice agent with streaming latency (a later phase).
- Moving money, changing bank details, or any irreversible financial action by the AI alone.
- Integration with real Paytm production systems.

Proposed success metrics (targets to validate in the prototype)

Metric	Definition	Proposed target
Autonomous resolution rate	Tickets closed with no human touch / all tickets	50-60% of in-scope tickets
Time to resolution	Voice note received to merchant confirmed fixed or booked	Under 3 minutes (vs. hours via queue)
Escalation quality	Escalations that arrive with diagnosis + summary	100%
Safe-action rate	Actions that stayed within policy tier	100% (hard guardrail)
Language coverage	Languages supported end to end	2 for prototype, 5+ roadmap

5. User journey (example)

Merchant voice note (Gujarati/Hinglish): "Kal ka settlement nahi aaya, aur speaker bhi band hai."

Step	What happens	Component
1. Capture	Merchant records a voice note (web recorder in demo; WhatsApp or call in roadmap)	Channel layer
2. Understand	Speech to text with language auto-detection; extract intents: settlement_delay + device_offline	STT + LLM
3. Gather context	Fetch merchant profile, last settlement, device heartbeat and error code	Agent tools
4. Decide	Classify cause (e.g., device offline due to network; settlement in bank-side processing window) and pick action tier	LLM + policy engine
5. Act	Tier 1: trigger remote restart/re-sync. Tier 2: propose technician visit, wait for one-click human approval. Tier 3: escalate with summary	Tools + approval queue
6. Reply	Compose a short reply in the merchant's language and speak it back; log every step	LLM + TTS + audit log

6. Functional requirements

ID	Requirement	Priority
F1	Accept a voice note (up to 30 s per clip) and return a transcript with detected language	P0
F2	Extract intent(s) and entities (merchant, device, date) with a confidence score	P0
F3	Agent calls tools: get_merchant, get_settlement_status, get_device_health, restart_device, create_technician_ticket, escalate_to_human, send_reply	P0
F4	Deterministic policy engine decides what the agent may do alone vs. with approval vs. never	P0
F5	Spoken reply in the merchant's language, plus a text copy	P0
F6	Human agent console: approval queue, escalation inbox, and a decision trace for each ticket	P0
F7	Confirm-before-act: when STT confidence is low or the action is not trivially reversible, ask the merchant to confirm	P1
F8	Immutable audit log of inputs, tool calls, decisions and outputs	P1

ID	Requirement	Priority
F9	Second language and WhatsApp channel	P1
F10	Proactive alerts (detect device offline before the merchant complains)	P2

Autonomy tiers and guardrails

Tier	Examples	Rule
1 - Auto	Status lookup, remote device restart/re-sync, reply with information	AI acts alone; logged
2 - Approve	Technician dispatch, retry of a delayed settlement request	AI prepares; human one-click approval
3 - Human only	Anything touching money movement, KYC, bank details, fraud suspicion, or low confidence	AI escalates with summary; never acts

The LLM proposes; the **policy engine (plain code, not the model)** authorises. This keeps the system predictable and is a key trust argument for the judges.

7. The voice feature: how it works and how hard it is

> **Verdict: moderate, not too tough.** Speech-to-text and text-to-speech are API calls. A working voice-note loop is realistic in 1-2 days. The hard parts are accuracy handling (numbers, code-mixed speech, noise) and latency for live calls. The plan below avoids the live-call problem for Round 2.

Pipeline

Stage	What it does	Option (verified from vendor docs)
Capture	Record audio in the browser (MediaRecorder) and upload as WAV	Web demo page; later WhatsApp Cloud API / Twilio
STT	Transcribe and detect language; handle Hindi-English mixing	Sarvam speech-to-text (Saaras v3): 22 Indian languages incl. Gujarati, auto language detection, code-mixed audio. REST API for clips under 30 s; WebSocket streaming for real time (16 kHz WAV/PCM)
Understand + act	Intent, tool calls, decision	Any tool-calling LLM (e.g., Claude, Gemini); policy engine in code
TTS	Speak the reply in the merchant's language	Sarvam Bulbul: 11 Indian languages incl. Gujarati; REST text limit of 1,500 chars (v2) / 2,500 chars (v3), so keep replies short
Fallback	If confidence is low, ask the merchant to repeat/confirm, or escalate	Own logic

Alternatives if Sarvam is unavailable: Google Cloud Speech-to-Text/Text-to-Speech, Azure Speech, or open-source Whisper for STT. Check current language coverage and pricing before committing; vendor details change.

Scoping for Round 2 (so it is achievable)

- **In scope:** asynchronous voice notes (clips under 30 s), Gujarati + Hindi/Hinglish, spoken + text reply. This needs no streaming and tolerates 3-6 s latency.
- **Out of scope:** live full-duplex phone calls, interruption handling, on-device Soundbox voice input (not verified that the device supports it; use the Soundbox only to announce results if at all).
- **Accuracy safeguards:** read back critical details ("Aapka kal ka settlement, 24 October, sahi?"), confirm before any action above Tier 1, and fall back to a human on low confidence.
- **Testing:** record 20-30 sample clips (quiet, noisy shop, code-mixed) and report transcript accuracy honestly in the demo.

8. System architecture and tech stack

Layer	Choice	Why
Frontend	Next.js/React: merchant voice page + agent console (approval queue, decision trace)	Fast to build; the trace view is your demo's hero screen
Backend / API	Python FastAPI (async)	Clean tool-calling orchestration; good SDK support
Agent loop	LLM with native tool calling; simple custom loop (LangGraph optional)	Fewer moving parts than a heavy framework
Policy engine	Rules in code (tier mapping, thresholds, allowed-tool lists)	Deterministic and auditable
Voice	Sarvam STT + TTS (see section 7)	Indic language and code-mixed support
Data	PostgreSQL (merchants, devices, settlements, tickets, audit log); Redis queue for approvals	Simple, standard

Layer	Choice	Why
Mock services	Small FastAPI service that simulates Paytm device/settlement APIs with seeded synthetic data	Real APIs are not accessible
Deploy	Docker; Render/Railway for API, Vercel for frontend	Free-tier friendly

Flow: Merchant audio -> Channel API -> STT -> Agent (LLM) <-> Tools (mock Paytm APIs) -> Policy engine -> {auto-act | approval queue | escalation} -> TTS -> Merchant. Every hop writes to the audit log.

9. Implementation plan

Phase	Deliverable	Effort (est.)
Now (before 27 Oct 11:59 PM IST)	Stage 1 pitch deck: 7 required sections + demo mock + architecture diagram	1 day
R2 - Phase 1	Mock Paytm services + seeded data; tool functions; policy engine	1-2 days
R2 - Phase 2	Agent loop with text input; decision trace stored in DB	1-2 days
R2 - Phase 3	Voice: record -> STT -> agent -> TTS; Gujarati + Hindi; confidence/confirm logic	1-2 days
R2 - Phase 4	Agent console (approvals, escalations, trace); polish; scripted demo with 3 scenarios	1-2 days
Finals prep	Demo rehearsal, backup recording, metrics slide from test runs	1 day

Effort estimates are rough planning guesses for a small team and depend on the Round 2 timeline, which should be confirmed with the organisers.

Demo scenarios (build these three)

- **Tier 1:** Soundbox offline; agent diagnoses a network drop, restarts remotely, confirms by voice.
- **Tier 2:** Hardware fault suspected; agent proposes a technician visit; human approves in one click.
- **Tier 3:** Merchant mentions an unauthorised transaction; agent refuses to act and escalates with a full summary.

10. Risks and mitigations

Risk	Mitigation
STT errors on noisy or code-mixed speech	Confidence threshold, read-back confirmation, human fallback; report measured accuracy honestly
LLM takes a wrong action	Policy engine authorises all actions; money/KYC/fraud are Tier 3 only
No access to real Paytm data	Mock services with realistic synthetic data; state this clearly in the pitch
Vendor API limits or outages	Abstract STT/TTS behind an interface; keep a fallback provider and pre-recorded demo
Scope creep before finals	Lock the three demo scenarios; defer proactive alerts and live calls
Privacy of voice and merchant data	Store minimal data, mask identifiers, document retention; synthetic data in demo

Appendix: mapping to the 7 required pitch-deck sections

Build the deck in this order. Aim for 8-10 slides total (the 7 required sections plus one demo/mock slide).

#	Required section	What to put on the slide	Draw from
1	Title	Name, one-line tagline ("Support that speaks the merchant's language and gets the job done"), Track 3, team	Cover
2	Problem statement	One merchant story (silent Soundbox, missing settlement), why voice/vernacular matters, 1-2 sourced figures	Sec. 2
3	Proposed solution	The 6-step journey as one flow diagram; the three autonomy tiers	Sec. 5, 6
4	Tech stack	Architecture diagram (channel, STT, agent, policy engine, tools, TTS, DB); name the tools	Sec. 7, 8
5	USP	Voice-first in Indian languages; acts rather than answers; risk-tiered autonomy with human handoff; full audit trail	Sec. 6, 7
6	Impact and benefits	Proposed metrics (label as targets): resolution rate, time to resolution, escalation quality, merchant downtime reduced	Sec. 4
7	Business model	Cost saved per ticket, merchant retention and activity, upsell path (e.g., value-added services), scalable across Paytm's merchant base	Sec. 1, 4
+	Demo slide (extra)	Screenshot of the agent decision trace and a sample voice exchange	Sec. 5, 9

Slide tips: one idea per slide; diagrams over paragraphs; label every number as target or sourced; state the mock-data assumption openly (judges respect honesty); end with the Round 2 plan so they see it is buildable.